

Aravind Kannappan

New York City, NY
1-408-591-5449
aravinds.kannappan@gmail.com
[LinkedIn](#) | [GitHub](#) | [Website](#)
Open To Relocation
US Citizen

EDUCATION

New York University
MS in Applied Statistics
Specialization in Machine Learning
Aug 2024 – Expected Aug 2026

Baylor University
BS in Statistics
Minor in Biology
Aug 2020 – Aug 2024

TECHNICAL SKILLS

Languages:
Python, TypeScript, SQL, R, C++

AI/ML Frameworks:
LangChain, LangGraph, PyTorch, Hugging Face, vLLM, Scikit-learn, RAG, Prompt Engineering, Tensorflow, MCP

Frontend: React, REST APIs

Cloud & DevOps:
AWS (S3, EC2, Bedrock), GCP, Docker, Kubernetes, CI/CD, Git

COURSEWORK

- NLP, Deep Learning, Data Engineering, Distributed Systems, Big Data, Bayesian Methods, Biostatistics
- Data Structures & Algorithms, Linear Algebra, Calculus, Database Systems, System Design, Operating Systems

TECHNICAL WRITING [BLOG](#)

- Wrote posts on optimization, generalization, and probabilistic modeling from first principles.
- Built reproducible Python notebooks, training scripts, and benchmarks.
- Studied failure modes (leakage, shift, calibration) and proposed fixes.

PUBLICATIONS

- Multiple Myeloma Gene Signatures (The Oncologist, 2024)**
Built statistical models in Python for mutation, pathway, and survival analysis, identifying gene signatures. [Link](#)
- Agentic AI Systems for Clinical Reasoning (Book Chapter, 2026)**
Developed frameworks for integrating clinical & social data to improve decision-making and patient-centered care. [Link](#)

SUMMARY

AI/ML Engineer with production experience designing LLM-powered systems, RAG pipelines, and multi-agent orchestration frameworks across healthcare, sports analytics, and AI safety research. Deployed microservices on AWS and GCP with CI/CD automation, HIPAA-aligned security, and ETL pipelines processing large scale datasets. Strong foundation in ML systems; comfortable driving full-stack AI delivery in fast-moving environments.

WORK EXPERIENCE

Research Engineer Fellow Jun 2026 – Present
Eleuther AI | Summer of Open AI Research *Remote*

- Applying mechanistic interpretability via sparse autoencoders on residual stream activations and linear probes to localize emergence of galaxy morphology across transformer layers
- Computing mutual k-NN alignment across foundation model embedding spaces on astronomical data to test PRH; evaluating cross-model convergence via cosine similarity and rank correlation.
- Implemented activation patching and feature ablation for causal interventions on SAE-learned feature dictionaries; built SLURM multi-GPU pipelines with mixed precision and gradient checkpointing for large-scale feature extraction.
- Trained crosscoder comparisons to study shared galaxy decomposition features; evaluated inter-model agreement via probing classifiers on held-out benchmarks.
- Built layer-wise feature extraction pipeline over galaxy embeddings with indexed FAISS; applied KL-divergence shift detection to identify representation emergence thresholds across model depth.

Machine Learning Engineer Feb 2026 – May 2026
Supervised Program for Alignment Research (SPAR) *Remote*

- Investigated AI safety properties of frontier LLMs using Bayesian methods to model AI consciousness; designed experiments across 3 models using Python and PyTorch for downstream analysis.
- Fine-tuned LLMs to generate 6 semantically equivalent question variants per indicator; measured cross-variant output distributions to stress-test alignment stability, reducing variance by 30%.
- Developed mathematical models characterizing ground truth drift in LLM judgment across model generations; analyzed how scaling and RLHF-driven training affect model behavior.
- Built statistical analysis over response distributions including Likert histograms, cross-model EDA, and semantic analysis of chain-of-thought justifications across 50+ safety-relevant indicators.
- Open-sourced full evaluation framework on GitHub for reproducible AI safety research; designed for longitudinal alignment drift tracking and integration with human expert survey data.

Software Engineer, AI/ML [Website](#) | [GitHub](#) Sep 2025 – Apr 2026
Replays AI | AI-Powered Immersive Post-Game Recaps *New York, NY*

- Designed ingestion pipeline writing 5M+ structured play-by-play events from the ESPN API into PostgreSQL as system of record; stored raw video referenced by pointer, cutting compute by 40%.
- Enforced schema validation, player disambiguation, and deduplication using Python; anchored OpenCV video segmentation before CV inference, enabling accurate highlight alignment.
- Orchestrated 3 parallel agentic inference pipelines via asyncio spanning event feature extraction, Claude Vision play classification (14 labels, confidence-scored), and reducing inference by 35%.
- Decoupled storage and inference from the frontend with FastAPI REST endpoints; implemented Redis recap caching with permanent TTL and fast-path rankings with 5-minute TTL, sustaining sub-second response times on cache hits.
- Built React 18 and TypeScript frontend with Vite and TanStack Query rendering personalized game feeds and fan-perspective AI recaps, reducing load time by 28% via component memoization.

Founder and ML Engineer [Website](#) | [GitHub](#) Jul 2023 – Jul 2025
Synthure | Mult-Agent Clinical Intelligence *New York, NY*

- Built production RAG pipeline with semantic and BM25 keyword search over 1.43M ICD-10 embeddings evaluated via offline benchmarks achieving MRR@5 of 0.91 and denial prediction.
- Engineered agentic orchestration infrastructure with observability gates enforcing NER confidence thresholds, schema validation, and deduplication as reliability checkpoints
- Implemented MCP constrained generation with output filtering and post-generation citation validation eliminating fabricated clinical facts across 60K+ production records.
- Managed prompt engineering at scale with templated chain-of-thought patterns and few-shot optimization across claim adjudication and entity extraction; reduced policy errors by 94%.
- Established evaluation methodology via held-out MTSamples benchmarks achieving 94.2% NER accuracy; deployed FastAPI with Redis caching and JWT HIPAA sustaining 1.8s p95 latency.

PROJECTS

CSCI 2271 Computer Vision [Github](#)

- Trained diffusion world model on 737K+ frames with VAE, CNN reward model and PPO agents

RobustSight: Advancing AI Safety and Alignment [Github](#)

- AI Safety framework investigating the intersection of adversarial robustness and interpretability